ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals
MATLAB Examples

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Topic 3: Digital Filter Fundamentals – Pole Zero / Filter response

Example 1

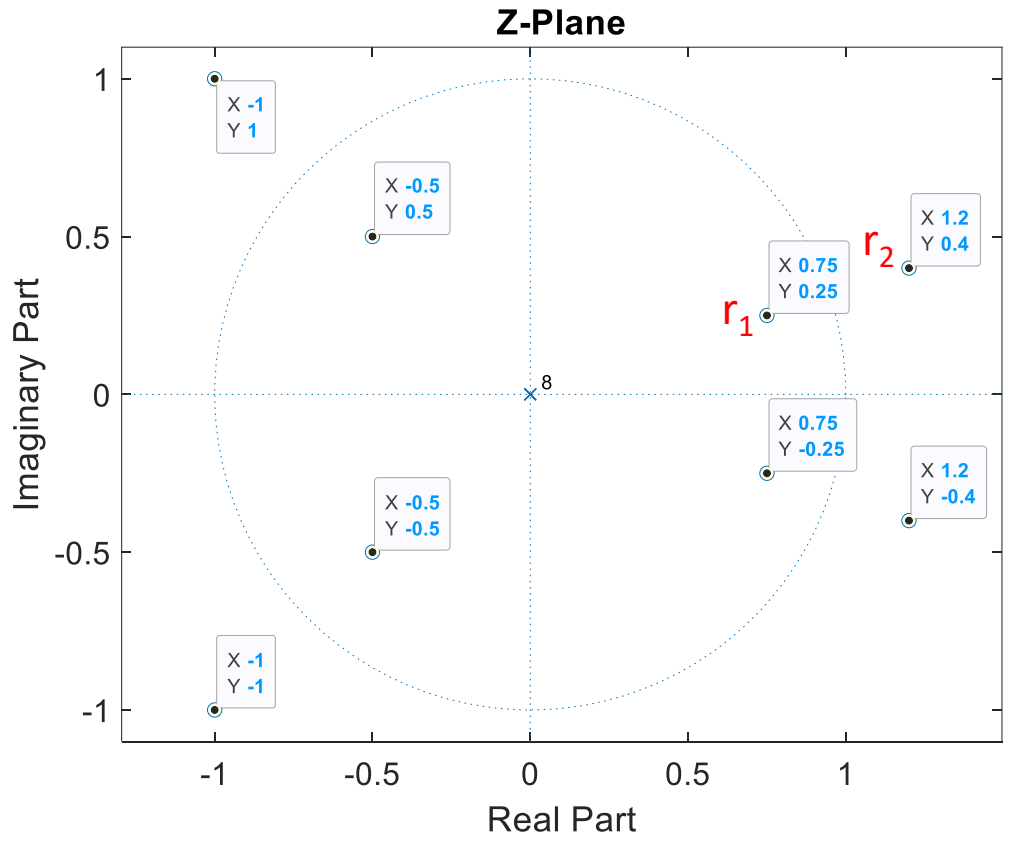
Consider the filter:

$$H(z) = \frac{z^8 - 0.9z^7 - 1.375z^6 - 0.975z^5 + 4.8125z^4 - 0.975z^3 - 1.375z^2 - 0.9z^1 + 1}{z^8}$$

Write a MATLAB code to:

a. Find the poles and zeros of this filter H(z) and plot them on the z-plane
b. Verify that the zeros of this filter are reciprocal to each other.
c. Plot the phase response of this filter to identify its special property
d. Find the impulse response of this filter
e. The input of this filter is x[n] = [1,3,5,7,9,11,9,7,5,3,1] for n=0,1,2,…10. Find the output of the filter

```
% part a
b = [1 -0.9 -1.375 -0.975 4.8125 -0.975 -1.375 -0.9 1]; % numerator coeffs
a= 1; % denominator coeffs
figure
zplane(b,a);
% part b
num_roots= roots(b); % Zeros of H(z)
disp(num_roots)
```



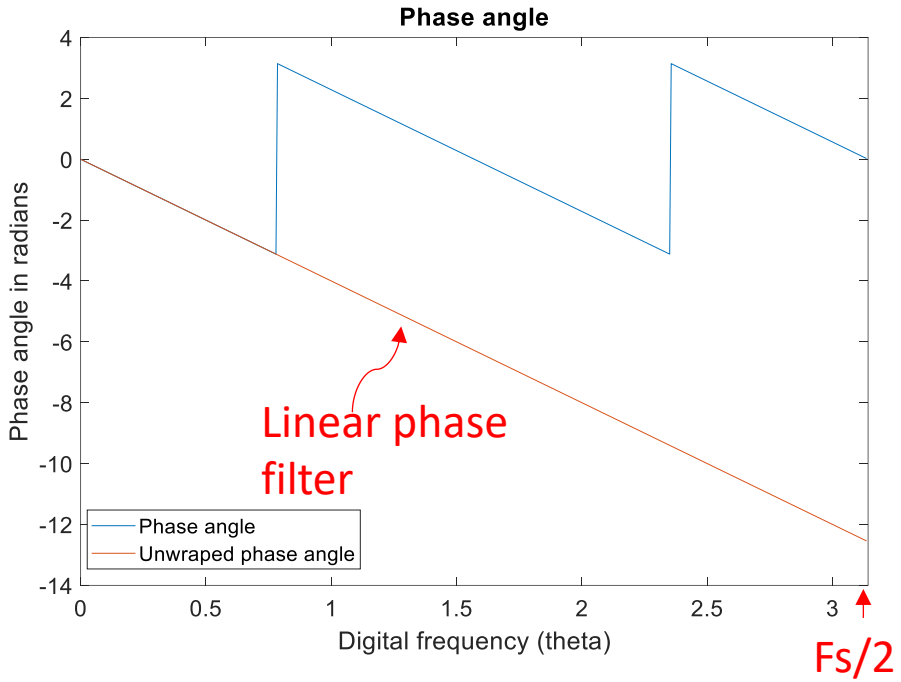
Roots r_1 and r_2 are reciprocal to each other and the same relationship holds for other roots

Roots (Complex conjigate) :

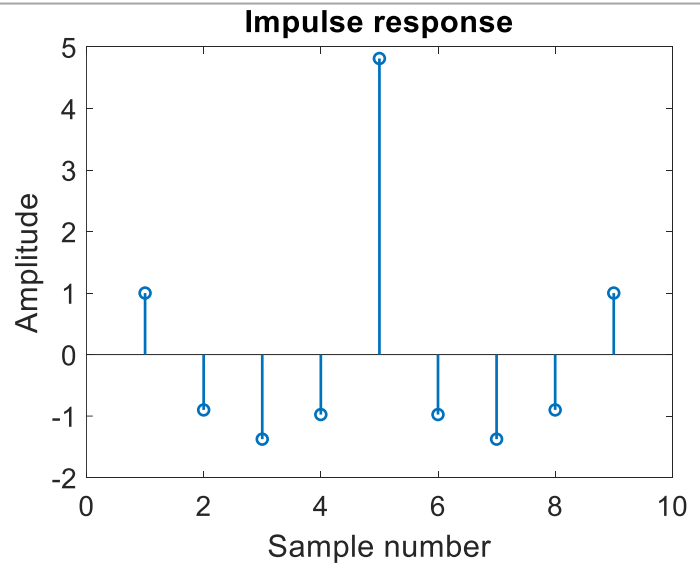
-1.0 + 1.0i, -1.0- 1.0i,
-0.5 + 0.5i, -0.5 - 0.5i,
1.2 + 0.4i, 1.2 - 0.4i,
0.8 + 0.3i, 0.8 - 0.3i

Topic 3: Digital Filter Fundamentals – Pole Zero / Filter response

```
% part c
[H,w]=freqz(b,a); % frequerncy response calculation
figure
plot(w, angle(H)); % plot the phase angle
hold on
plot(w, unwrap(angle(H))); % plot the unwraped phase angle
xlim([0,pi]); % x axis upto half the sampling frequency
xlabel('Digital frequency (theta)');
ylabel('Phase angle in radians');
title("Phase angle");
```




```
% part d
h=impz(b,a);
figure
stem(h);
title('Impulse response');
xlabel('Sample number');
ylabel('Amplitude');
```




```
% part e
x= [1,3,5,7,9,11,9,7,5,3,1];
y= conv(x, h); % h obtained in pard d
figure
stem(y); xlabel('Sample number n'); ylabel('y[n]');
xlim([0,10]);
```

y = [1.0000 2.1000 0.9250 -2.6000 -2.2875 1.8625 -0.3375 -1.2125 3.5125 13.1375 3.5125 -1.2125 -0.3375 1.8625 -2.2875 -2.6000 0.9250 2.1000 1.0000]

length of y = length of x + length of h - 1

Topic 3: Digital Filter Fundamentals – Phase and Group Delay

Example 2

A digital filter has eight poles at the origin and eight zeros as given below

Zeros	$0.75 \pm j\,0.25$; $-0.5 \pm j\,0.5$; $1.2 \pm j\,0.4$; $-1 \pm j$

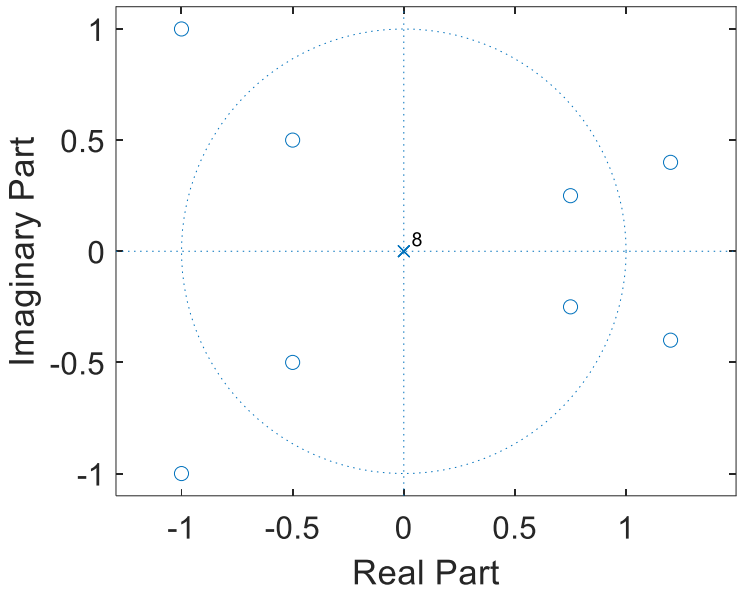
Write a MATLAB code to:

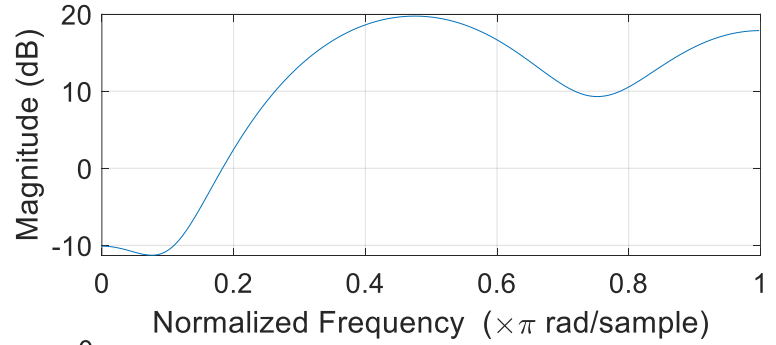
a. Plot the pole-zero plane (z-plane) and a frequency response of this filter. Comment on its phase response and the location of zeros.

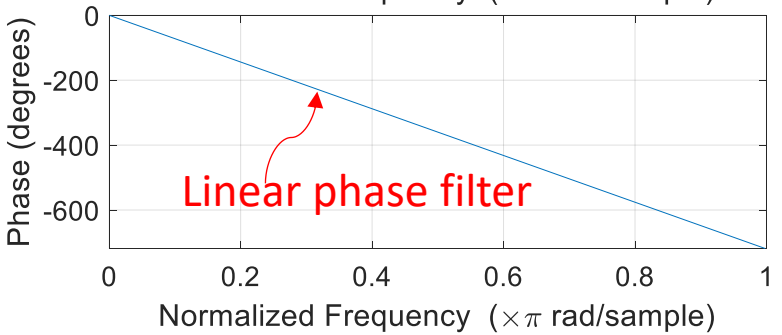
b. Calculate the group delay of this filter

```
z = [0.75+j*0.25; 0.75-j*0.25; -0.5+j*0.5; -0.5-j*0.5; 1.2-j*0.4; 1.2+j*0.4; -1-j; -1+j];
b = poly(z); % provide the coefficients of the filter
a= 1;

figure
zplane(b, a); % pole-zero plot
figure;
freqz(b,a); % plot the mag and phase response directly
```

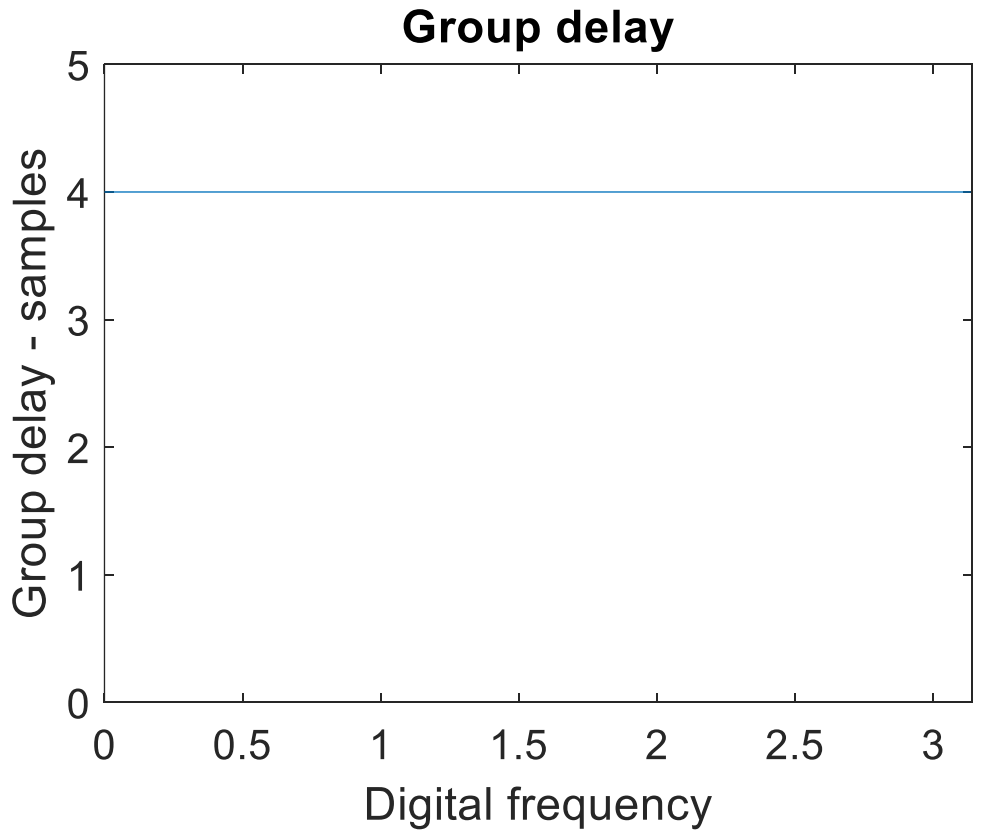






Topic 3: Digital Filter Fundamentals – Phase and Group Delay

```
% part b
[gD, w] = grpdelay(b); % calculate group delay
figure
plot(w, gD);
xlim ([0, pi]);
ylim([0,5]);
title ("Group delay");
xlabel("Digital frequency");
ylabel("Group delay - samples");
```



Topic 3: Digital Filter Fundamentals – Min/Max Phase Filters

Example 3

The digital filters A, B, and C have four poles at the origin and four zeros as shown on table below.

Write a MATLAB code to:

a. Plot the pole-zero on the z-plane and the frequency response (magnitude and phase respones) of each filter.
b. Are there any differences in their magnitude responses?
c. Comment on the phase lag of each of these filters.
d. Classify each of the above filters as either maximum phase, mixed phase or minimum phase filters

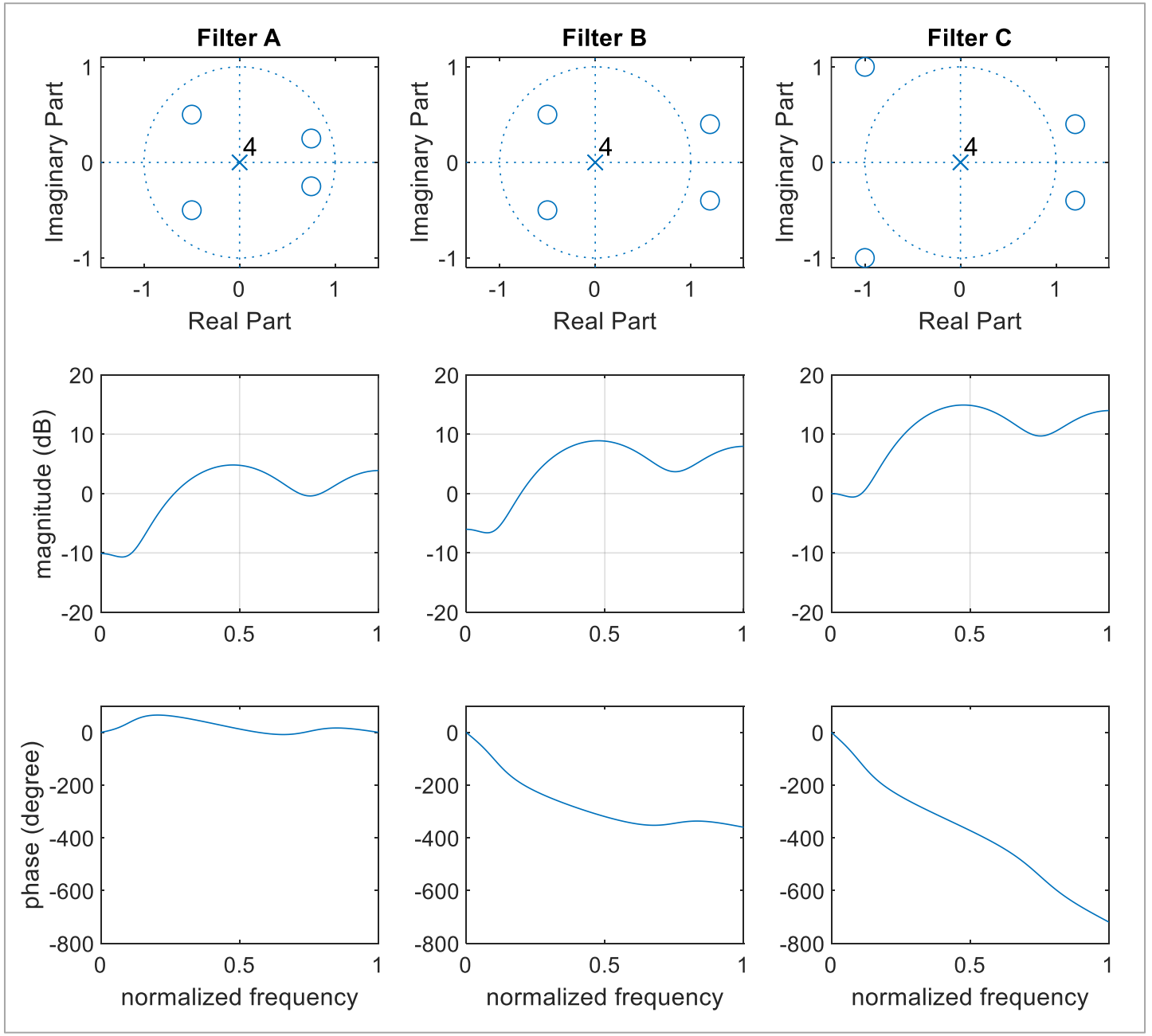
Note: Discuss your understanding of the above with your lab demonstrator

	Zeros Location
Filter A	0.75 ± j 0.25 ; -0.5 ± j 0.5
Filter B	1.2 ± j 0.4 ; -0.5 ± j 0.5
Filter C	1.2 ± j 0.4; -1 ± j

```
zeros_A= [0.75+j*0.25;0.75-j*0.25; -0.5+j*0.5; -0.5-j*0.5]
zeros_B = [1.2+j*0.4; 1.2-j*0.4; -0.5+j*0.5; -0.5-j*0.5];
zeros_C = [1.2+j*0.4; 1.2-j*0.4;-1-j;-1+j];
b_A = poly(zeros_A); b_B = poly(zeros_B); b_C = poly(zeros_C);
a= 1;
[HA, w]=freqz(b_A,a); [HB, w] =freqz(b_B,a); [HC,w] =freqz(b_C,a );
% Ploting 3 z-planes, magnitude spectrum and phase spectrum
figure
subplot 331
zplane(b_A, a);
title('Filter A');
subplot 332
zplane(b_B, a);
title('Filter B');
subplot 333
zplane(b_C, a);
title('Filter C');
subplot 334
plot(w/w( end),20*log10(abs(HA)));
ylabel('magnitude (dB)');
grid on
```

Topic 3: Digital Filter Fundamentals – Min/Max Phase Filters

```
axis([0 1 -20 20])
subplot 335
plot(w/w(end),20*log10(abs(HB)))
axis([0 1 -20 20]);
grid on
subplot 336
plot(w/w(end),20*log10(abs(HC)))
axis([0 1 -20 20]); grid on
subplot 337
plot(w/w(end),unwrap(angle(HA))/pi*180)
xlabel('normalized frequency')
ylabel('phase (degree)')
axis([0 1 -800 100])
subplot 338
plot(w/w(end),unwrap(angle(HB))/pi*180)
xlabel('normalized frequency')
axis([0 1 -800 100])
subplot 339
plot(w/w(end),unwrap(angle(HC))/pi*180)
xlabel('normalized frequency')
axis([0 1 -800 100])
```



ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals
MATLAB Exercises

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Exercises – Digital Filter Fundamentals (Topic 3)

Exercise 1 - Magnitude and Phase Responses

A digital filter is given by the following transfer function

$$H(z) = \frac{3-2z+z^2}{1-2z+3z^2}$$

Write a MATLAB code to:

a. Plot the magnitude response of this digital filter.
b. Find the zeros and poles of $H(z)$ and plot the poles-zeros on the z-plane.
c. Compute and plot the phase response of the filter .

Note: Comment on the magnitude response, phase response and the pole-zero locations of this filter and discuss your results with your lab demonstrator.

Exercise 2: Cut-off Frequency and Bandwidth

A digital filter is given by the following transfer function

$$H(z) = \frac{0.2}{1-0.8z^{-1}}$$

I. Find the cut-off frequency and bandwidth of the above digital filter.
II. Write a MATLAB code to:
    a. Compute and plot the magnitude response of the system and automatically compute the cut-off frequency and bandwidth of the filter.
    b. Compare the results that you obtained in part a above with (I).

Exercises – Digital Filter Fundamentals (Topic 3)

Exercise 1 - Magnitude and Phase Responses

A digital filter is given by the following transfer function

$$H(z) = \frac{3-2z+z^2}{1-2z+3z^2}$$

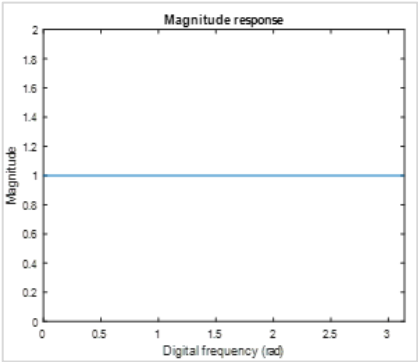
Write a MATLAB code to:

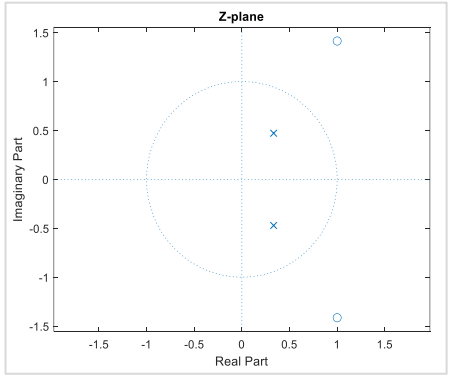
a. Plot the magnitude response of this digital filter.
b. Find the zeros and poles of $H(z)$ and plot the poles-zeros on the z-plane.
c. Compute and plot the phase response of the filter .

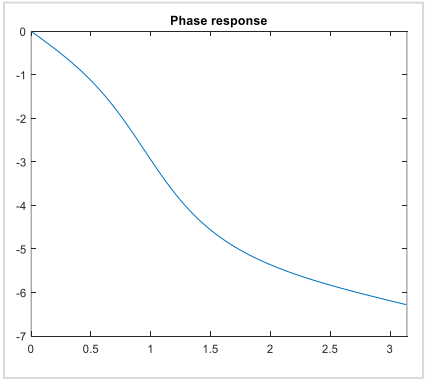
```
% part a
a=[3, -2, 1]; % denominator coefficients
b=[1, -2, 3]; % numerator coefficients
[H, w]=freqz(b,a); % frequency response
figure
% |H| is constant therefore it is a all-pass filter
plot(w, abs(H));
xlabel('Digital frequency (rad)');  ylabel('Magnitude');
title('Magnitude response');  axis([0 pi 0 2]);

% part b
zeros= roots(b);   poles= roots(a);
% Verify that each pole is a reciprocal of a zero
zplane(zeros, poles);
% zplane(b,a);  % alternatively

% part c
figure
plot(w, unwrap(angle(H)));  xlim([0,pi]); title('Phase response');
```







Exercises – Digital Filter Fundamentals (Topic 3)

Exercise 2: Cut-off Frequency and Bandwidth

A digital filter is given by the following transfer function

$$H(z) = \frac{0.2}{1-0.8z^{-1}}$$

I. Find the cut-off frequency and bandwidth of the above digital filter.

II. Write a MATLAB code to:

   a. Compute and plot the magnitude response of the system and automatically compute the cut-off frequency and bandwidth of the filter.

   b. Compare the results that you obtained in part a above with (I).

```
b = 0.2; % H(z)'s numerator coefficients
a = [1 -0.8]; % H(z)'s denominator coefficients
[H, w] = freqz(b,a,2048); % compute frequency response
Hm = 20*log10(abs(H)); % compute magnitude response in dB
wc = w(max(find(Hm>-3))); % find the cut-off frequency
BW = wc; % bandwidth of filter
figure
plot(w, Hm); hold on
% drawing vertical dotted line
plot(wc*ones(10,1),linspace(-20, Hm(find(w==wc)),10),'--');
hold on
% drawing horizontal dotted line at -3 dB
plot(linspace(0,wc,5),-3*ones(5,1),'--');
axis([0, pi, -20, 0]); xlabel('Digital frequency (rad)');
ylabel('magnitude(dB)')
```

